

. A field $\mathcal{F}$ is a set equipped with two binary operations: $+$ & $\cdot$, such that:

① closed: $\forall x, y \in \mathcal{F}, x + y \in \mathcal{F}$.

② Associativity: $\forall x, y, z \in \mathcal{F}, (x + y) + z = x + (y + z)$.

③ Commutativity: $\forall x, y \in \mathcal{F}, x + y = y + x$.

④ Identity: $\exists 1 \neq 0$, s.t. $\forall x \in \mathcal{F}, x + 0 = x$ & $x \cdot 1 = x$.

⑤ Inverse:

$\forall x \in \mathcal{F}, \exists x_I \in \mathcal{F}$ s.t. $x + x_I = 0$	$\forall x \in \mathcal{F}, x \neq 0, \exists x^{-1} \in \mathcal{F}$ s.t. $x \cdot x^{-1} = 1$
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⑥ Distributivity (interaction of $+$ & $\cdot$): $\forall x, y, z \in \mathcal{F}, x \cdot (y + z) = x \cdot y + x \cdot z$.

* Derived Properties :

① The field's 0 is unique.

② The additive inverse of every element in F is unique.

③ $\forall x \in F, x \cdot 0 = 0$.

. ①-③ were proved in the lecture.

④ The field's 1 is unique :

Proof: . let a & $b \in F$ be multiplicative iden.

. $a \neq b$ (assume identity isn't unique)

. $\forall x \in F$:

. $x \cdot a = x$ (Def. of identity)
. let $x = b$.

$\therefore b \cdot a = b \rightarrow \textcircled{\text{I}}$.

. $x \cdot b = x$ (Def. of identity)
. let $x = a$.

$\therefore a \cdot b = a$.

. using commutativity, $a \cdot b = b \cdot a$.

$\therefore b \cdot a = a \rightarrow \textcircled{\text{II}}$.

. From $\textcircled{\text{I}}$ & $\textcircled{\text{II}}$: $a = b$ which contradicts our assumption that $a \neq b$.

∴ The field's 1 is unique.

⑤ The multi. inverse of every element in $\mathbb{F}$ is unique ($x \neq 0$) :

Proof : . Consider any $x \in \mathbb{F}$ ($x \neq 0$).

. let a & $b \in \mathbb{F}$ be multi. inverses of x .

. Assume $a \neq b$.

. $x \cdot a = 1$ (Def. of multi. inverse).
. $x \cdot b = 1$ (Ditto).

. $\underbrace{(x \cdot a)}_1 \cdot b = 1 \cdot b = b \rightarrow \textcircled{\text{I}}$.

. $(x \cdot a) \cdot b = x \cdot (a \cdot b)$ (Assoc.)
 $= x \cdot (b \cdot a)$ (Comm.)
 $= \underbrace{(x \cdot b)}_1 \cdot a$ (Assoc.)
 $= 1 \cdot \frac{1}{a} = a \rightarrow \textcircled{\text{II}}$.

. From $\textcircled{\text{I}}$ & $\textcircled{\text{II}}$: $a = b$, a contradiction.

so x^{-1} is unique.

$$\textcircled{6} \forall x \in \mathcal{F}, \quad x_I = \underbrace{1_I}_{\substack{\text{Add inverse} \\ \text{of the field's} \\ 1}} \cdot x$$

This is a formula to get x_I .

Proof: • let $1_I \cdot x = y \in \mathcal{F}$ (we don't know yet the value of $1_I \cdot x$).

$$\bullet x + y = \underbrace{1 \cdot x}_{\text{identity}} + 1_I \cdot x$$

$$= (1 + 1_I) \cdot x \quad (\text{Distributivity})$$

$$= 0 \cdot x \quad (1 + 1_I = 0)$$

$$= x \cdot 0 \quad (\text{comm.}),$$

$$= 0 \quad (\text{property } \textcircled{3})$$

$$\text{so } x + 1_I \cdot x = 0$$

$$\text{so } \underbrace{x_I + x + 1_I \cdot x}_{=0} = x_I + 0 = x_I$$

$$\text{so } 0 + 1_I \cdot x = x_I$$

$$\text{so } 1_I \cdot x = x_I$$

e.g. $\circ \mathbb{F} = \mathbb{R}$, $(3)_\mathbb{F} = \underbrace{(-1)}_{(1)_\mathbb{F}} * 3$.

* Let $\mathbb{F} = \{(x, y) \mid x, y \in \mathbb{R}\}$.

$(x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$.
 $\uparrow \quad \quad \quad \uparrow$
 $\mathbb{F} \text{ add.} \quad \quad \quad \mathbb{R} \text{ add.}$

$(x_1, x_2) \odot (y_1, y_2) = (x_1 y_1, -x_2 y_2, x_1 y_2 + x_2 y_1)$
 $\uparrow \quad \quad \quad \uparrow \quad \quad \quad \downarrow$
 $\mathbb{F} \text{ mult.} \quad \quad \quad \mathbb{R} \text{ mult.} \quad \quad \quad (1_\mathbb{F}) \cdot (x_2 y_2)$
 $\uparrow$
 $\text{add. inverse of } \mathbb{R}'\text{'s } 1$

* Show that $\mathbb{F}$ equipped with $\oplus$ & $\odot$ is a field, what's $0_\mathbb{F}$, $1_\mathbb{F}$, $x_\mathbb{F}^{-1}$ & $x_{1_\mathbb{F}}$?

. Consider addition:

* Does $(x_1 + y_1, x_2 + y_2) \in \mathbb{F}$?

. For any $(z_1, z_2) \in \mathbb{F}$, z_1 & z_2 must $\in \mathbb{R}$.

. $x_1 + y_1 \in \mathbb{R}$ (real addition is closed).

Similarly $x_2 + y_2 \in \mathbb{R}$.

$\omega(x_1 + y_1, x_2 + y_2) \in \mathbb{F}$, is closed $\checkmark$.

* Does $(x_1, x_2) \oplus (y_1, y_2) \oplus (z_1, z_2)$.

$$= (x_1, x_2) \oplus ((y_1, y_2) \oplus (z_1, z_2))?$$

Consider L.H.S

$$(x_1 + y_1, x_2 + y_2) \oplus (z_1, z_2) =$$

$$(x_1 + y_1 + z_1, x_2 + y_2 + z_2)$$

R.H.S.

$$(x_1, x_2) \oplus (y_1 + z_1, y_2 + z_2) =$$

$$(x_1 + y_1 + z_1, x_2 + y_2 + z_2)$$

= L.H.S, is ASSOC. $\checkmark$.

* Does $(x_1, x_2) \oplus (y_1, y_2) = (y_1, y_2) \oplus (x_1, x_2)$.

$$\text{L.H.S} = (x_1 + y_1, x_2 + y_2)$$

$$\text{R.H.S} = (y_1 + x_1, y_2 + x_2)$$

$\hookrightarrow x_1 + y_1$ (read + is assoc)

$$= (x_1 + y_1, x_2 + y_2) = \text{L.H.S.}$$

is Comm. $\checkmark$.

* what's the field's $0_{\mathbb{F}}$?

• By definition $\underbrace{(x_1, x_2)}_{\alpha} \oplus \underbrace{(0, 0)}_{0_{\mathbb{F}}} = \alpha$.

• $\forall \alpha \in \mathbb{F}$.

$$\circ (x_1 + 0_1, x_2 + 0_2) = (x_1, x_2).$$

$$\circ x_1 + 0_1 = x_1 \xleftarrow{+ (x_1)_I \in \mathbb{R}}.$$

Assoc. $\downarrow$

$$\circ 0_1 + x_1 \neq x_1 = 0 \xrightarrow{0 \text{ (real } \delta)}$$

$$\circ 0_1 + 0 = 0, \circ 0_1 = 0.$$

$\downarrow \in \mathbb{R}$

• Similarly, $0_2 = 0$.

$$\circ 0_{\mathbb{F}} = (0, 0), \circ \text{Identity } \checkmark.$$

* what's α_I for $\alpha \in \mathbb{F}$?

• By definition $\underbrace{(x_1, x_2)}_{\alpha} \oplus \underbrace{(y_1, y_2)}_{\alpha_I} = 0_{\mathbb{F}}$.

$$\circ (x_1 + y_1, x_2 + y_2) = (0, 0).$$

$$\circ x_1 + y_1 = 0.$$

$\xrightarrow{\text{Add}} (x_1)_I \in \mathbb{R}$

$$\circ -x_1 + x_1 + y_1 = 0 + y_1 = -x_1.$$

$\xrightarrow{0 \in \mathbb{R}}$

$$\circ y_1 = -x_1.$$

• Similarly, $y_2 = -x_2$.

• α_I of $(X_1, X_2) = (-X_1, -X_2)$.

• Add. inverse ✓.

↳ we'll consider this again later

• Consider multiplication:

* Does $(X_1 Y_1 \mp X_2 Y_2, X_1 Y_2 + X_2 Y_1) \in \mathbb{R}$?

↳ let's see if this definition of multiplication is valid.

• $X, Y \in \mathbb{R}$ (real multi. is closed).

• $(\mp 1) \cdot X_2 Y_2 \in \mathbb{R}$. (")

• Addition of two real numbers is real.

• $X_1 Y_1 \mp X_2 Y_2 \in \mathbb{R}$, similarly $X_1 Y_2 + X_2 Y_1 \in \mathbb{R}$.

• $(X_1 Y_1 \mp X_2 Y_2, X_1 Y_2 + X_2 Y_1) \in \mathbb{R}$, • closed ✓.

* Does $(X_1, X_2) \odot (Y_1, Y_2) = (Y_1, Y_2) \odot (X_1, X_2)$?

• L.H.S = $(X_1 Y_1 \mp X_2 Y_2, X_1 Y_2 + X_2 Y_1)$.

• R.H.S = $(Y_1 X_1 \mp Y_2 X_2, Y_1 X_2 + Y_2 X_1)$.

• $Y_1 X_1 = X_1 Y_1$ (Real '•' is comm.).

$$\cdot \gamma_2 x_2 = x_2 \gamma_2 \quad (\quad // \quad).$$

$$\circledast \gamma_1 x_1 \mp \gamma_2 x_2 = x_1 \gamma_1 \mp x_2 \gamma_2.$$

$$\cdot \gamma_1 x_2 = x_2 \gamma_1 \text{ \& } \gamma_2 x_1 = x_1 \gamma_2 \quad (//).$$

$$\begin{aligned} \circledast \gamma_1 x_2 + \gamma_2 x_1 &= x_2 \gamma_1 + x_1 \gamma_2 \\ &= x_1 \gamma_2 + x_2 \gamma_1. \end{aligned}$$

(Real '+' is comm.).

$$\circledast \text{L.H.S} = \text{R.H.S} , \circledast \text{Comm. } \checkmark$$

$$\begin{aligned} * \text{ Does } & ((x_1, x_2) \odot (\gamma_1, \gamma_2)) \odot (z_1, z_2) \\ &= (x_1, x_2) \odot ((\gamma_1, \gamma_2) \odot (z_1, z_2)) ? \end{aligned}$$

$$\begin{aligned} \bullet \text{ L.H.S } & \circledast \left(\overbrace{x_1 \gamma_1 \mp x_2 \gamma_2}^{u_1}, \overbrace{x_1 \gamma_2 + x_2 \gamma_1}^{u_2} \right) \odot \\ & (z_1, z_2) \\ &= \underbrace{(u_1 z_1 \mp u_2 z_2)}_a, \underbrace{(u_1 z_2 + u_2 z_1)}_b. \end{aligned}$$

$$\begin{aligned} \bullet \text{ R.H.S } & \circledast (x_1, x_2) \odot \underbrace{(\gamma_1 z_1 \mp \gamma_2 z_2)}_{v_1}, \gamma_1 z_2 \\ & + \gamma_2 z_1). \\ & \quad \downarrow^c \quad \quad \downarrow^d \downarrow^d \\ &= (x_1 v_1 \mp x_2 v_2, x_1 v_2 + x_2 v_1). \end{aligned}$$

$$\cdot a = c ?$$

$$\begin{aligned} \cdot a &= (x_1 y_1 + x_2 y_2) z_1 + (x_1 y_2 + x_2 y_1) z_2 \\ &= x_1 y_1 z_1 + x_2 y_2 z_1 + x_1 y_2 z_2 + x_2 y_1 z_2. \end{aligned}$$

(Distributive property of the real field).

$$\begin{aligned} \cdot c &= x_1 (y_1 z_1 + y_2 z_2) + x_2 (y_1 z_2 + y_2 z_1) \\ &= x_1 y_1 z_1 + x_1 y_2 z_2 + x_2 y_1 z_2 + x_2 y_2 z_1. \end{aligned}$$

By real '+' assoc. & comm., applied multiple times

$$\cdot c = x_1 y_1 z_1 + x_2 y_2 z_1 + x_1 y_2 z_2 + x_2 y_1 z_2.$$

$$\text{So } a = c.$$

$$\cdot b = d ?$$

$$\begin{aligned} \cdot b &= (x_1 y_1 + x_2 y_2) z_2 + (x_1 y_2 + x_2 y_1) z_1 \\ &= x_1 y_1 z_2 + x_2 y_2 z_2 + x_1 y_2 z_1 + x_2 y_1 z_1. \end{aligned}$$

$$\begin{aligned} \cdot d &= x_1 (y_1 z_2 + y_2 z_1) + x_2 (y_1 z_1 + y_2 z_2) \\ &= x_1 y_1 z_2 + x_1 y_2 z_1 + x_2 y_1 z_1 + x_2 y_2 z_2. \end{aligned}$$

. By assoc. & comm. of real '+' :

$$. b = d .$$

∴ Comm. ✓.

* What's $1_{\mathbb{F}}$ of the field?

. By definition $\underbrace{(x_1, x_2)}_{\alpha} \odot \underbrace{(1_1, 1_2)}_{1_{\mathbb{F}}} = \alpha$
 $\forall \alpha \in \mathbb{F}.$

$$\circ \circ (x_1 \cdot 1_1 \mp x_2 1_2, x_1 1_2 + x_2 1_1) = (x_1, x_2).$$

$$\circ \circ x_1 \cdot 1_1 \mp x_2 1_2 = x_1 \longrightarrow \textcircled{\text{I}}$$

$$. x_1 1_2 + x_2 1_1 = x_2 \longrightarrow \textcircled{\text{II}} .$$

$$. \text{From } \textcircled{\text{II}} : x_1 1_2 = x_2 - x_2 1_1 .$$

$$\circ \circ x_1 1_2 = x_2 (1 - 1_1)$$

$\hookrightarrow$ Real 1.

. Assume $x_1 \neq 0$ (will handle this later).

$$\circ \circ 1_2 = \frac{x_2}{x_1} (1 - 1_1).$$

. Sub. in $\textcircled{\text{I}}$:

$$. x_1 \cdot 1_1 \mp \frac{x_2^2}{x_1} (1 - 1_1) = x_1 .$$

$$\circ \circ x_1 \cdot 1_1 \mp \frac{x_2^2}{x_1} \pm \frac{x_2^2}{x_1} 1_1 = x_1 .$$

$$. (x_1 \pm x_2^2/x_1) \cdot 1_1 = x_1 \pm x_2^2/x_1 .$$

• We want this to hold $\forall x_1, x_2$, so
 $1_1 = 1 \in \text{real } 1$.

$$\text{so } 1_2 = \frac{x_2}{x_1} \underbrace{(1/1)}_{\rightarrow 0} = 0.$$

$$\text{so } 1_{\mathbb{R}} = (1, 0).$$

• Identity ✓.

• Recall: $\alpha = (x_1, x_2)$, $\alpha_I = (-x_1, x_2)$.

$$\bullet (1_{\mathbb{R}})_I = (-1, 0).$$

$$\begin{aligned} \bullet (1_{\mathbb{R}})_I \odot \alpha &= (-1, 0) \odot (x_1, x_2) \\ &= (-x_1, -x_2). \end{aligned}$$

• Formula for additive inverse.

$$\begin{aligned} * \text{ Now, let } x_1 = 0, \text{ so } (\mp x_2 1_2, x_2 1_1) &= (x_1, x_2) \\ &= (0, x_2). \end{aligned}$$

$$\text{so } \mp x_2 1_2 = 0 \quad \forall x_2, \text{ so } 1_2 = 0.$$

$$\bullet x_2 1_1 = x_2 \quad \forall x_2, \text{ so } 1_1 = 1, \text{ so } 1_{\mathbb{R}} = (1, 0).$$

* What's α^{-1} s.t. $\forall \alpha \in \mathbb{R}, \alpha \neq 0$

$$\bullet \alpha \odot \alpha^{-1} = 1_{\mathbb{R}}.$$

$$\cdot (x_1, x_2) \odot (y_1, y_2) = (1, 0).$$

$$\circ \alpha \neq 0, \circ \circ (x_1, x_2) \neq (0, 0).$$

$$\cdot \text{Req. } (x_1 y_1 \mp x_2 y_2, x_1 y_2 + x_2 y_1) = (1, 0).$$

$$\cdot x_1 y_1 \mp x_2 y_2 = 1 \rightarrow \textcircled{\text{I}}.$$

$$\cdot x_1 y_2 + x_2 y_1 = 0 \rightarrow \textcircled{\text{II}}.$$

$$\cdot \text{From } \textcircled{\text{II}} \circ y_2 = \frac{-x_2}{x_1} y_1 \text{ (let } x_1 \neq 0 \text{ for now)}.$$

$$\cdot \text{Sub. in } \textcircled{\text{I}} \circ$$

$$\cdot x_1 y_1 \pm \frac{x_2^2}{x_1} y_1 = 1.$$

$$\circ \circ \left(\frac{x_1^2 \pm x_2^2}{x_1} \right) y_1 = 1.$$

$$\circ \circ y_1 = \frac{x_1}{x_1^2 \pm x_2^2}, \quad y_2 = \frac{-x_2}{x_1^2 \pm x_2^2}.$$

• Now consider the two different fields we had :

$$\cdot \textcircled{1} \circ \circ (x_1, x_2) \odot (y_1, y_2) = (x_1 y_1 - x_2 y_2, x_1 y_2 + x_2 y_1)$$

$$\cdot y_1 = x_1 / (x_1^2 + x_2^2), \quad y_2 = -x_2 / (x_1^2 + x_2^2).$$

∴ we assumed $x_1 \neq 0$, ∴ $x_1^2 + x_2^2 \neq 0$.

∴ γ_1, γ_2 are valid.

$$\text{∴ } \alpha^{-1} = \left(\frac{x_1}{x_1^2 + x_2^2}, \frac{-x_2}{x_1^2 + x_2^2} \right).$$

• Let $x_1 = 0$, ∴ Req. $(0, x_2) \odot (\gamma_1, \gamma_2) = (1, 0)$.

• $x_2 \neq 0$, o.w., $(x_1, x_2) = (0, 0) = 0_{\mathbb{F}}$

$$\text{∴ } (-x_2 \gamma_2, x_2 \gamma_1) = (1, 0).$$

$$\text{∴ } -x_2 \gamma_2 = 1, \text{ \& } \gamma_2 = -\frac{1}{x_2} = \frac{-x_2}{x_2^2 + \underbrace{x_1^2}_{=0}}.$$

$$\text{• } x_2 \gamma_1 = 0, \text{ ∴ } \gamma_1 = 0 = \frac{x_1}{x_1^2 + x_2^2}.$$

$$\text{∴ } \alpha^{-1} = \left(\frac{x_1}{x_1^2 + x_2^2}, \frac{-x_2}{x_1^2 + x_2^2} \right) \text{ in general.}$$

∴ $\forall \alpha \in \mathbb{F}_1, \alpha \neq 0, \exists \alpha^{-1} \in \mathbb{F}_1 \text{ s.t.}$

$$\alpha \odot \alpha^{-1} = 1_{\mathbb{F}_1}.$$

• This is the complex field.

$$\text{• } j = (0, 1).$$

$$\text{• } j \odot j = (0, 1) \odot (0, 1) = (-1, 0).$$

$$\text{• let } \alpha_i = (x_1, x_2).$$

- $\alpha_2 = (x_1, -x_2)$.

- Then $\alpha_2 = \alpha_1^*$.

- $\alpha_1 \odot \alpha_2 = \alpha_1 \odot \alpha_1^* = |\alpha_1|^2 = x_1^2 + x_2^2$.

- That's why $\alpha^{-1} = (x_1, -x_2) \odot \left(\frac{1}{x_1^2 + x_2^2}, 0 \right)$
 $= \alpha^* \odot \left(\frac{1}{|\alpha|^2}, 0 \right)$.

* Now consider the second field $\mathbb{F}_2$

- $(x_1, x_2) \odot (y_1, y_2) = (x_1 y_1 + x_2 y_2, x_1 y_2 + x_2 y_1)$
 $= (1, 0)$.

- $y_1 = \frac{x_1}{x_1^2 - x_2^2}$, if $x_1 = x_2$, then y_1

doesn't exist.

- E.g. $(1, 1) \overset{\alpha}{\odot} (x_1, x_2) \overset{\alpha^{-1}}{=} (1, 0)$.

- $(x_1 + x_2, x_1 + x_2) = (1, 0)$.

- $x_1 + x_2 = 1$ & $x_1 + x_2 = 0$.

• This system of equations doesn't have a solution.

- $\exists \alpha \in \mathbb{F}_2, \alpha \neq 0$ s.t. $\nexists \alpha^{-1}$ satisfying $\alpha \odot \alpha^{-1} = 1_{\mathbb{F}_2}$.

∴ This isn't a field.

* we check distributivity for $\mathbb{F}_1$ only now (the complex field) since it's the only candidate left:

$$\begin{aligned} * \text{Does } (x_1, x_2) \odot ((y_1, y_2) \oplus (z_1, z_2)) \\ = (x_1, x_2) \odot (y_1, y_2) \oplus (x_1, x_2) \\ \odot (z_1, z_2). \end{aligned}$$

$$\begin{aligned} \text{L.H.S.} &= (x_1, x_2) \odot (y_1 + z_1, y_2 + z_2) \\ &= (\underbrace{x_1(y_1 + z_1) - x_2(y_2 + z_2)}_a, \\ &\quad \underbrace{x_1(y_2 + z_2) + x_2(y_1 + z_1)}_b). \end{aligned}$$

$$\text{R.H.S.} =$$

$$\begin{aligned} &(x_1 y_1 - x_2 y_2, x_1 y_2 + x_2 y_1) \oplus \\ &(x_1 z_1 - x_2 z_2, x_1 z_2 + x_2 z_1) \\ &= (\underbrace{x_1 y_1 - x_2 y_2 + x_1 z_1 - x_2 z_2}_c, \\ &\quad \underbrace{x_1 y_2 + x_2 y_1 + x_1 z_2 + x_2 z_1}_d). \end{aligned}$$

$$\text{. } a = c?$$

. Yes, by real comm., assoc. & distr.

. $b=d$? Yes, //.

∞ Distributivity ✓.

∞ $\mathbb{F}_1$ is a valid field.

* A vector space V , over a field $\mathbb{F}$, is a non-empty set, equipped with:

. '+' : $V \times V \rightarrow V$. (Closed under add.)

. '·' : $\mathbb{F} \times V \rightarrow V$. (Closed under scalar mult.).

These binary operations satisfy:

. A) '+' :

① Associative : $\forall x, y, z \in V$

$$(x+y)+z = x+(y+z).$$

② Commutative : $\forall x, y \in V$

$$x+y = y+x.$$

③ Identity : $\exists 0_V \in V$ s.t.

$$\forall x \in V, x+0_V = x.$$

④ Inverse : $\forall x \in V \exists x_I$ s.t.
 $x + x_I = 0_V$.

B) ' $\cdot$ ' :

⑤ Associative : $\forall \alpha, \beta \in F, x \in V$
 $(\alpha \cdot \beta) \cdot x = \alpha \cdot (\beta \cdot x)$.

⑥ Identity : let 1_F be the field's
1, then $\forall x \in V, 1_F \cdot x = x$.

C) ' $+$ ' & ' $\cdot$ ' interaction (Distributivity)

: $\forall \alpha, \beta \in F, x, y \in V$

⑦ $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$.

⑧ $(\alpha + \beta) \cdot \vec{v} = \alpha \cdot \vec{v} + \beta \cdot \vec{v}$
 $\swarrow \quad \searrow$
 $\leftarrow F_{add} \quad \quad \quad \leftarrow V_{add}$

*Derived properties :

① 0_V is unique :

Proof : let $u, v \in V$ be zeros & $u \neq v$.

$$\forall x \in V :$$

$$x + u = x. \quad (\text{Definition of } 0_V)$$

$$\circledast V + u = V \rightarrow \textcircled{I}.$$

$$x + v = x. \quad (//).$$

$$\circledast u + v = u$$

$$\parallel \\ v + u = u \rightarrow \textcircled{II}. \quad (\text{Comm.}).$$

$\circledast$ From $\textcircled{I}$ & $\textcircled{II}$ $\circledast$ $V = u$, $\circledast$ contradiction.

$\circledast$ 0_V is unique.

② $\forall x \in V$, x_I is unique :

proof: Consider any $x \in V$.

Let $v, u \in V$ be add. inverse of x .

Assume $u \neq v$.

By def. $\circledast$ $x + v = 0_V$ & $x + u = 0_V$.

$$(x + u) + v = 0_V + v = v \rightarrow \textcircled{I}.$$

Assoc. $\hookrightarrow$

$$x + (u + v) \stackrel{\text{Comm.}}{=} x + (v + u)$$

$$\stackrel{\text{Assoc.}}{=} (x + v) + u$$

$$= 0_V + u = u \rightarrow \textcircled{II}.$$

From $\textcircled{I}$ & $\textcircled{II}$ $\circledast$ $u = v$, $\circledast$ contradiction.